

Zero-Shot Cross-Lingual Recognition of Sign Language Handshapes

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Abstract

Sign language processing advances rapidly for high-resource languages such as American Sign Language (ASL), yet most of the world’s sign languages lack the phonological annotations new methods require. We present the first zero-shot cross-lingual framework for handshape recognition, transferring from ASL to Catalan Sign Language (LSC). Our approach leverages the decomposition of handshapes into five phonological features — selected fingers, flexion, spread, thumb position, and thumb contact — shared across both languages, to decode LSC handshapes from predicted features via a composite phonological distance metric. We evaluate three architectures (MLP, SL-GCN, SHuBERT) trained on two ASL corpora (PopSign, Sem-Lex) against a 37-handshape, single-signer LSC benchmark. Zero-shot transfer proves viable once recording-format disparities are harmonized, reaching 80.0% phonological feature accuracy and 54.5% expected handshape accuracy. Phonological decomposition thus offers a bridge for extending sign language technologies to low-resource languages without any target-language video training labels.

1 Introduction

While over 70 million deaf people globally use over 200 distinct sign languages (SLs) (World Federation of the Deaf, 2026), most remain low-resource (Bragg et al., 2019). Annotated corpora and modeling efforts are heavily concentrated on a few high-resource languages like American Sign Language (ASL). In contrast, Catalan Sign Language (LSC) lacks the annotated resources needed to drive modern recognition systems, beyond a gloss-annotated dataset (Institut d’Estudis Catalans, 2025) and comprehensive grammar (Quer and Barberà, 2020).

Sign Language Processing (SLP) is bottlenecked by annotation costs: labels are costly and require

fluent experts, and gloss-level supervision does not transfer across distinct lexicons. Phonological features offer a compelling solution. They provide a linguistically grounded label set that bridges languages. Because handshapes decompose into core phonological features—*selected fingers*, *finger flexion*, *spread*, *thumb position*, and *thumb contact* (Brentari, 1998)—a model trained on a high-resource language could theoretically recognize unseen target-language handshapes by recombining these shared phonological features.

While monolingual phonological recognition is well established (Kezar et al., 2023; Carbo and Nalisnick, 2025; Gueuwou et al., 2025; Inoue et al., 2026), and prior cross-lingual modeling relies on target-language annotations (Tornay et al., 2020; Bilge et al., 2024), strictly zero-shot cross-lingual phonological recognition remains unexplored.

We address this gap by transferring from ASL to LSC. Our contributions are as follows.

- The first zero-shot cross-lingual handshape recognition framework, evaluated across three architectural paradigms (static MLP, SL-GCN (Jiang et al., 2021), and SHuBERT (Gueuwou et al., 2025)), using two ASL corpora (PopSign (Starner et al., 2023) and Sem-Lex (Kezar et al., 2023)).
- A formal phonological alignment scheme mapping Navarrete-González’s (2020) LSC handshapes into the ASL-LEX 2.0 (Sehyr et al., 2021) phonological feature space.
- A distance-based decoding protocol and expected-accuracy metric designed to handle mappings from phonological features to target-language handshapes.
- Empirical evidence that transfer is viable—achieving up to 80.0% phonological feature accuracy and 54.5% expected handshape accuracy—but only after harmonizing recording-format disparities.

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2 Related work

Our contribution bridges five lines of work that structure this section. Cross-lingual phonological transfer relies on three foundations: a shared feature inventory (*phonology of sign languages*), robust sign encoding models (*sign language processing*), and annotated resources (*phonology-curated datasets*). While their convergence enables monolingual *phonology recognition*, extending this to *cross-lingual phonological recognition* remains an open frontier—where we position this work.

Phonology of Sign Languages Stokoe (1960) and Battison (1978) established that Manual Features of signs decompose into contrastive sub-lexical units: handshape, palm orientation, location, and movement. Subsequent frameworks like Brentari’s (1998) Prosodic Model refined these into a hierarchical feature geometry, decomposing handshapes into finer-grained features such as selected fingers, finger flexion, spread, and thumb position; and decomposing movement into path and location components. Modern datasets like ASL-LEX (Caselli et al., 2017; Sehyr et al., 2021) code phonology precisely at this granular level. Crucially, LSC relies on these same sub-handshape parameters (Quer and Barberà, 2020; Navarrete-González, 2020). This structural overlap, stemming from their shared lineage in the *Francosign* family (Wittmann, 1991), motivates the cross-linguistic transfer pursued in this work.

Sign Language Processing Early SLP relied on gloss-based pipelines constrained by comparatively small annotated datasets like RWTH-PHOENIX-Weather dataset (Forster et al., 2012, 2014; Camgoz et al., 2018, 2020). Recent advances have shifted the paradigm toward gloss-free translation (Zhou et al., 2023; Lin et al., 2023; Müller et al., 2023), mapping video or pose sequences directly to text, but critically depend on robust visual features. To address this, recent work leverages self-supervised pre-training on large, unannotated video corpora like YouTube-ASL (Uthus et al., 2023) and YouTube-SL-25 (Tanzer and Zhang, 2025). Models like SHuBERT (Gueuwou et al., 2025) use these datasets to learn generalized spatio-temporal representations without costly gloss annotations.

Phonology-Curated Datasets Interpreting fine-grained sub-lexical structures requires dedicated phonological resources. Several curated datasets have emerged, though available resources remain

overwhelmingly restricted to ASL: ASL-LEX 2.0 (Sehyr et al., 2021) catalogs properties for over 2,700 glosses; Sem-Lex (Kezar et al., 2023) aligns deaf signers’ isolated productions to ASL-LEX 2.0; and PopSign (Starner et al., 2023; Chow et al., 2023) offers game-collected sign recordings linked to these phonological features. We refer the reader to Section 3 for further dataset details.

Phonology Recognition Early work focused on isolated feature extraction, with milestones like DeepHand (Koller et al., 2016) targeting handshape recognition. To enable deeper phonological understanding, subsequent systems have leveraged on the aforementioned curated datasets to capture complex spatial and temporal variations. For example, Kezar et al. (2023) demonstrated that phonology recognition is an effective auxiliary target for Isolated Sign Language Recognition (ISLR), showing that an SL-GCN (Jiang et al., 2021) trained on Sem-Lex achieves 85% average accuracy across the 16 ASL-LEX phonological feature types. For handshape recognition, the Handshape-GNN model (Carbo and Nalisnick, 2025)—trained for the PopSign Kaggle challenge (Chow et al., 2023)—separates temporal sign dynamics from static hand configurations, benchmarking effectively against a multilayer perceptron (MLP) baseline. Similarly, the self-supervised SHuBERT (Gueuwou et al., 2025) achieved state-of-the-art ISLR accuracy on Sem-Lex when fine-tuned, suggesting that its learned representations implicitly encode sub-lexical phonological structure.

Cross-lingual Phonological Recognition Despite successes with rich, language-specific training data, the aforementioned models have not been evaluated on lower resourced SLs in a cross-lingual setup. Prior cross-lingual approaches adapt phonological subunits on target-language data (Tornay et al., 2020) or recognize novel signs from a few labeled target-language examples (Bilge et al., 2024). However, target-language annotation is a prohibitively expensive bottleneck for underdocumented sign languages. A zero-shot approach bypasses this by leveraging shared cross-linguistic phonological structures. To our knowledge, zero-shot cross-lingual sign phonology prediction remains unexplored. We address this gap, demonstrating successful zero-shot transfer and establishing the first baseline for this task.

3 Datasets

To evaluate cross-lingual transfer across diverse capture conditions and signers, our framework combines three ASL resources with a target LSC benchmark. Table 1 synthesizes their primary characteristics.

ASL-LEX 2.0 (Sehyr et al., 2021) serves as the ASL lexical-phonological reference dictionary, providing over 2,700 glosses with detailed phonological annotations. While the dataset tracks 16 phonological features, our handshape prediction pipeline employs the 5 phonological features strictly related to the handshape (*selected fingers*, *flexion*, *spread*, *thumb position*, and *thumb contact*) plus handshape class. Crucially, the combination of these 5 features is not always a strict 1-to-1 mapping. As detailed by Sehyr et al. (2021), phonological feature combinations can group non-contrastive variants together or fail to differentiate contrastive handshapes that vary in unselected finger flexion (see Section 6).

Sem-Lex (Kezar et al., 2023) connects ASL-LEX 2.0 annotations to a large-scale corpus of over 84,000 webcam video instances representing 3,149 signs recorded remotely by 41 Deaf signers. The dataset combines ASL-LEX citation signs, free-text responses, and SignBank (Hochgesang et al., 2019) entries. Due to reproducibility issues with the official pre-extracted poses, we re-extracted all poses directly from the raw video source.

PopSign (Starner et al., 2023) introduces in-the-wild mobile recordings collected from 47 ASL learners, comprising $\sim 175k$ videos covering 250 isolated glosses balanced across all three dataset splits. We utilize the raw video release of the game split rather than the landmark-only Kaggle release (Chow et al., 2023), as RGB source videos are required for hybrid multimodal architectures (e.g., SHuBERT) that process pixel features alongside pose sequences.

LSC Benchmark¹ was accessed as a pre-release subset to evaluate zero-shot cross-lingual transfer. It comprises 37 target handshapes performed by a native Deaf LSC signer. Structured according to SignHub’s LSC finger configuration (Navarrete-González, 2020), each handshape contains two videos: (1) a target handshape demonstration, and (2) an example sign.

¹A pre-release subset was shared with us prior to publication; redistribution and visual sample display are restricted by the dataset creators.

Dataset	Lang.	#Hs	#Glosses	#Videos	#Signers	Rec. Type
ASL-LEX 2.0 [†]	ASL	58	2,723	2,723	–	Studio
Sem-Lex	ASL	58	3,149	84,568	41 Deaf	Webcam
PopSign	ASL	37	250	175,022	47 Learners	Smartphone
LSC Benchmark	LSC	37	–	74	1 Deaf	Studio

Table 1: Overview and comparison of datasets evaluated in our experiments. Lang.: sign language; #Hs.: number of unique handshape classes; Rec. Type: video recording environment or device. [†]ASL-LEX 2.0 was used strictly as a lexical mapping reference; its video data was not utilized in our experiments.

4 Methodology

4.1 Model Architectures

To benchmark phonological feature prediction and cross-lingual transferability, we evaluate three representative architectural paradigms: a lightweight pose-based MLP baseline, a spatio-temporal graph neural network (SL-GCN) (Jiang et al., 2021), and a multi-modal pre-trained transformer (SHuBERT) (Gueuwou et al., 2025). Each architecture is trained and evaluated across two distinct ASL source datasets spanning different capture modalities: Sem-Lex and PopSign. To ensure complete reproducibility, we publicly release our model implementations, hyperparameter configurations, and evaluation pipelines, together with trained model checkpoints.²

4.1.1 Baseline MLP

Following the static baseline architecture introduced by Carbo and Nalisnick (2025), we employ a Multi-Layer Perceptron (MLP) with 3 layers of dimensions $63 \rightarrow 256 \rightarrow 256 \rightarrow N_{\text{classes}}$. The model operates strictly on static spatial features, removing the temporal dimension by extracting a single representative frame per sign sequence corresponding to minimal hand motion. Input features consist of raw 3D hand landmarks ($21 \times 3 = 63$ features) extracted via MediaPipe (Lugaresi et al., 2019). The network is trained from scratch on Sem-Lex and PopSign independently to predict multi-label phonological feature targets. This model serves as a computationally efficient reference to compare with more complex temporal and multi-modal architectures.

4.1.2 SL-GCN

To capture dynamic spatio-temporal hand movements and finger interactions, we employ the

²<https://marcelgranero.github.io/cross-lingual-handshapes>

Sign Language Graph Convolutional Network (SL-GCN) (Jiang et al., 2021), which served as the primary baseline for the Sem-Lex benchmark (Kezar et al., 2023). SL-GCN models pose sequences as spatio-temporal graphs, applying graph convolutions across skeletal joints and temporal convolutions across consecutive frames. Following Kezar et al.’s (2023), we first train the network on ISLR before fine-tuning for joint gloss and phonological feature prediction across both Sem-Lex and PopSign. To maintain a unified extraction pipeline across both datasets (Section 3) and achieve optimal pose fidelity, all input pose sequences are re-extracted via MediaPipe (Lugaresi et al., 2019) directly from the raw video sources. The complete reproduction benchmarks and validation metrics are provided in Table 10 (Appendix C).

4.1.3 SHuBERT

To test whether large-scale video-pose pre-training enhances zero-shot cross-lingual transfer compared to purely pose-based models, we evaluate SHuBERT (Gueuwou et al., 2025). SHuBERT is a self-supervised multimodal transformer pre-trained on YouTube-ASL (Uthus et al., 2023) that fuses raw video pixels with pose sequences. We fine-tune the pre-trained encoder for phonological feature prediction on Sem-Lex and PopSign. Reproduction performance benchmarks against the original paper are detailed in Table 11 (Appendix C).

4.2 Zero-Shot Cross-Lingual Transfer

To evaluate zero-shot handshape recognition across languages, we map target LSC handshapes to five phonological features in ASL-LEX format. Models trained on ASL predict this shared phonological feature space, and we map the resulting predicted features to target LSC handshape classes with a distance-based metric. Note that this protocol is zero-shot with respect to target-language videos and video-level labels, but it does presuppose a dictionary-level target resource: a predefined LSC handshape inventory and its handshape-to-feature mapping (Appendix A).

4.2.1 Aligning LSC Handshapes to ASL-LEX Phonological Features

To bridge the structural differences between ASL-LEX 2.0 and the SignHub LSC annotations (SH-LSC), we establish a rule-based alignment that deterministically assigns the five ASL-LEX phonological features pertaining strictly to handshape

configuration —*selected fingers, finger flexion, spread, thumb position, and thumb contact*— to each LSC handshape (detailed in Appendices A and B). Specifically, Spread and Thumb Contact are mapped directly from SH-LSC to ASL-LEX categories without structural modifications. Finger flexion is mapped by harmonizing label conventions across schemes. For example, SH-LSC *Extended* is mapped to ASL-LEX *FullyOpen*. Additionally, because SH-LSC uses a single *curved* category, *curved* maps to a combined *Curved/Bent* class formed by joining ASL-LEX *Curved* and *Bent*. Regarding thumb phonology, SH-LSC treats the thumb as a standard selected finger, whereas ASL-LEX excludes the thumb from *Selected Fingers* unless it is the sole active digit. To align these paradigms, we map SH-LSC *Selected Fingers* onto ASL-LEX *Selected Fingers* and *Thumb Position*: if an SH-LSC handshape includes the thumb t^3 in its selected set, t is re-encoded as *Thumb Position = Open* and removed from *Selected Fingers*, unless t is the only selected digit (e.g., SH-LSC $\{t, i, m\} \rightarrow \text{ASL-LEX } \{i, m\} + \text{Open}$).

4.2.2 Phonological Handshape Matching and Evaluation

Phonological Representation Space Let $\mathcal{P}_{\text{sel}}, \mathcal{P}_{\text{flex}}, \mathcal{P}_{\text{sprd}}, \mathcal{P}_{\text{th_pos}}, \mathcal{P}_{\text{th_cnt}}$ be discrete categorical sets representing five articulatory features: *selected fingers, finger flexion, spread, thumb position, and thumb contact*, respectively. The phonological domain $\mathcal{P}$ is defined as their Cartesian product: $\mathcal{P} = \mathcal{P}_{\text{sel}} \times \mathcal{P}_{\text{flex}} \times \mathcal{P}_{\text{sprd}} \times \mathcal{P}_{\text{th_pos}} \times \mathcal{P}_{\text{th_cnt}}$. An arbitrary phonological configuration is thus represented as a 5-dimensional column vector $\mathbf{p} = [p_{\text{sel}}, p_{\text{flex}}, p_{\text{sprd}}, p_{\text{th_pos}}, p_{\text{th_cnt}}]^T \in \mathcal{P}$, where each component p_k takes a value from its corresponding feature set $\mathcal{P}_k$. Let $\mathcal{H} = \{h_1, h_2, \dots, h_C\}$ denote the set of $C = 37$ target LSC handshape classes. Each class $h \in \mathcal{H}$ is associated with a canonical phonological feature vector $\mathbf{p}^{(h)} \in \mathcal{P}$ (See Table 7). We define the set of valid LSC representations as $\mathcal{P}_{\text{LSC}} = \{\mathbf{p}^{(h)} \mid h \in \mathcal{H}\} \subset \mathcal{P}$. Notably, the handshape-to-feature mapping $h \mapsto \mathbf{p}^{(h)}$ is non-injective: two pairs⁴ among the 37 handshape

³Selected fingers are denoted as thumb t , index i , middle m , ring r , and pinkie p .

⁴7-flat_open with index+thumb-flat_open and 7-flat_closed with index+thumb-flat_closed coincide on phonological features. These handshapes differ on the unselected fingers flexion – closed or open for 7 and index+thumb respectively.

classes share identical feature vectors, yielding $|\mathcal{P}_{\text{LSC}}| = 35$ unique canonical feature configurations. Combinations in $\mathcal{P}$ that do not correspond to any valid LSC handshake form the set of unassociated phonological vectors $\mathcal{U} = \mathcal{P} \setminus \mathcal{P}_{\text{LSC}}$.

Inference and Constrained Decoding During inference, the ASL-trained models generate categorical predictions for the five phonological channels. Because ASL’s *selected fingers* and *finger flexion* features contain more classes than those in LSC (see Appendix B), we restrict each model’s predictions to valid LSC target classes. We achieve this by applying a target-space mask to the output logits prior to the argmax operation⁵, yielding a predicted feature vector $\hat{\mathbf{p}} = [\hat{p}_{\text{sel}}, \hat{p}_{\text{flex}}, \hat{p}_{\text{sprd}}, \hat{p}_{\text{th_pos}}, \hat{p}_{\text{th_cnt}}]^T \in \mathcal{P}$. However, since each feature is predicted by an independent classification head, nothing prevents the joint prediction $\hat{\mathbf{p}}$ from falling into $\mathcal{U}$.

Composite Phonological Distance Metric

Given a discrete predicted feature vector $\hat{\mathbf{p}} \in \mathcal{P}$ and a target handshake vector $\mathbf{p}^{(h)} \in \mathcal{P}_{\text{LSC}}$, we measure their dissimilarity using a composite distance function $d(\hat{\mathbf{p}}, \mathbf{p}^{(h)}) = \sum_{c \in \mathcal{C}} d_c(\hat{p}_c, p_c^{(h)})$ where $\mathcal{C} = \{\text{sel}, \text{flex}, \text{sprd}, \text{th_pos}, \text{th_cnt}\}$ represents the set of feature categories. The category distances d_c are defined as follows. First, d_{sel} is the Hamming distance between the two selected-finger representations, where a mismatch (present vs. absent) for any of the four individual fingers (i, m, r, p) adds 1 to the distance. Second, d_{flex} is defined over the four flexion values (*FullyOpen*, *Flat*, *Curved/Bent*, and *FullyClosed*), assigning a distance of 1 to every distinct pair except the extreme transition *FullyOpen* $\leftrightarrow$ *FullyClosed*, which costs 2. Finally, d_{sprd} , $d_{\text{th_pos}}$, and $d_{\text{th_cnt}}$ assign a distance of 0 for matching values⁶ and 1 for mismatches.

Candidate Selection and Evaluation The model outputs discrete categorical class predictions for each channel. Rather than requiring an exact vector match, we map the predicted feature vector $\hat{\mathbf{p}}$ to a set of candidate target handshapes $\hat{\mathcal{H}} \subset \mathcal{H}$ by retrieving all classes whose feature vectors minimize $d(\hat{\mathbf{p}}, \mathbf{p}^{(h)})$: $\hat{\mathcal{H}} = \arg\min_{h \in \mathcal{H}} d(\hat{\mathbf{p}}, \mathbf{p}^{(h)})$. Because multiple handshapes can tie for the minimum

Candidates (h)	Phonological Features ($\mathbf{p}^{(h)}$)	$d(\hat{\mathbf{p}}, \mathbf{p}^{(h)})$
Prediction ($\hat{\mathbf{p}} \in \mathcal{U}$)	$\{i, m, r\}$, FullyOpen, 1, Open, 0	–
8-extended	$\{i, m\}$, FullyOpen, 1, Open, 0	1
5-extended	$\{i, m, r, p\}$, FullyOpen, 1, Open, 0	1
3-extended	$\{i, m, r\}$, FullyOpen, 1, Closed , 0	1
2-extended	$\{i, m\}$, FullyOpen, 1, Closed , 0	2
4-extended	$\{i, m, r, p\}$, FullyOpen, 1, Closed , 0	2
5-curved-open	$\{i, m, r, p\}$, Curved/Bent , 1, Open, 0	2
8-curved-open	$\{i, m\}$, Curved/Bent , 1, Open, 0	2
B-extended	$\{i, m, r, p\}$, FullyOpen, 0 , Open, 0	2

Table 2: Candidate selection for a prediction in $\mathcal{U}$. Three target classes tie at minimum distance $d = 1$, forming $\hat{\mathcal{H}}$. Next-nearest classes ($d = 2$) are excluded. Mismatched features are in red.

distance, $\hat{\mathcal{H}}$ is a candidate set containing one or more class labels. We evaluate performance using *Expected Accuracy*. Let $h^* \in \mathcal{H}$ denote the true target class. Sample-level expected accuracy $A(h^*, \hat{\mathcal{H}})$ measures the probability of selecting h^* under a uniform random choice from $\hat{\mathcal{H}}$, defined as $A(h^*, \hat{\mathcal{H}}) = 1/|\hat{\mathcal{H}}|$ if $h^* \in \hat{\mathcal{H}}$, and 0 otherwise. Table 2 illustrates a prediction $\hat{\mathbf{p}} \in \mathcal{U}$ mapped to a candidate set $\hat{\mathcal{H}}$ of three equidistant target classes ($d = 1$)⁷. This instance scores $1/3$ if $h^* \in \hat{\mathcal{H}}$, and 0 otherwise.

4.3 Mitigating Recording-Format Disparities

The LSC benchmark consists of uniform 16:9 1080p full-body studio footage recorded at 50 fps. In contrast, the training corpora rely on in-the-wild recordings—webcam video for Sem-Lex and smartphone footage for PopSign—with both training sets having a median frame rate near 30 fps. To isolate cross-lingual errors from these recording-format disparities, we modify the LSC Benchmark clips to match the training capture geometries and subsample them to lower frame rates. To address spatial disparities, we extract a static bounding box centered on the median shoulder position for each target LSC clip, scaling it to match the training corpus medians. Specifically, the target normalized shoulder width and vertical y -position are aligned with Sem-Lex’s 4:3 landscape webcam format (denoted as webcam) and PopSign’s 3:4 portrait smartphone format (denoted as phone). Finally, to address temporal disparities, frame rate subsampling is applied to the LSC target clips after feature extraction using strides of 2 (25 fps) and 3 (16.7 fps). Integer strides avoid frame interpolation; 25 fps approaches the training corpora’s frame-rate medians, while 16.7 fps probes robustness below them. Note that

⁵We mask three *flexion* classes (*Crossed*, *Stacked*, and *NA*) and four *selected fingers* combinations (*imp*, *mr*, *mrp*, and *r*).

⁶We include the NA class as a distinct, valid state for Spread and Thumb contact.

⁷See [SignHub’s Finger Configuration](#).

harmonization uses no target-side supervision: the crop is computed independently for each clip from its own pose sequence together with source-corpus constants, without labels or any fitting across the test set, and is therefore applicable to any input video at inference time.

5 Results

In this section, we empirically evaluate the cross-lingual transferability of phonological representations from ASL to LSC. We first verify our baseline source-task reproductions and address pose extraction artifacts. Next, we evaluate zero-shot transfer performance on the LSC benchmark for the three model architectures (MLP, SL-GCN, and SHuBERT) and the two training corpora (Sem-Lex and PopSign). Finally, we present a series of ablations to isolate the key drivers of transfer performance, examining the impact of recording-format alignment, auxiliary supervision heads, and SHuBERT’s input channels and pretraining.

Source-Task Reproduction and Pose Correction Table 3 compares our reproduced performance on Sem-Lex against the metrics reported for SL-GCN (Kezar et al., 2023) and SHuBERT (Gueuwou et al., 2025). While our SHuBERT reproduction closely matches the published results, reaching 80.2% test macro Recall@1 versus the reported 79.4%, SL-GCN gloss recognition exhibits a noticeable gap. Training on the released Sem-Lex pose files yields 57.1% Top-1 accuracy, underperforming the reported 66.6% by 9.5 percentage points. However, when the model is trained with the poses we extracted from the source videos—instead of Sem-Lex released poses—and following the identical recipe, gloss top-1 accuracy increases to 72.2%, surpassing the published baseline rather than falling 9.5 points below it. Then, we fine-tune it for joint gloss and phonological feature prediction and improve the phonological feature average from 78.7% to 87.1%. The shortfall could be a property of the released data. Elsewhere in this paper the SL-GCN Sem-Lex rows refer to the model trained with re-extracted poses. Further reproduction details are provided in Appendix C.

Zero-shot transfer to LSC We train all three model architectures on PopSign and Sem-Lex using the full set of 16 phonological features. This choice maintains consistency with prior work and we hypothesize that additional feature targets pro-

Model	Gloss Top-1 (%) $\uparrow$			16-Feat. Ph. (%) $\uparrow$		
	Ref.	Ours	Δ	Ref.	Ours	Δ
SL-GCN (released poses)	66.6	57.1	-9.5	85.0	78.7	-6.3
SL-GCN (re-posed)	66.6	72.2	+5.6	85.0	87.1	+2.1
SHuBERT	—	—	—	79.4	80.2	+0.9

Table 3: Reproduction fidelity of published Sem-Lex-trained source models on the test set. Gloss Top-1: top-1 sign gloss classification accuracy; 16-Feat. Ph.: mean accuracy across all 16 phonological features. Ref. denotes published baselines from Kezar et al. (2023) for SL-GCN and Gueuwou et al. (2025) for SHuBERT. The Δ columns give absolute percentage point differences between our reproduction and the reference.

vide auxiliary supervision without degrading performance, as ablated later in this section. However, since our cross-lingual handshape recognition approach relies only on five handshape-related phonological features, we evaluate performance using the mean accuracy across these five phonological features (Ph.% $\uparrow$) and the expected handshape accuracy (Hs.% $\uparrow$). All models are evaluated across three random seeds both in-domain on the corresponding ASL test splits and zero-shot on LSC, reporting mean performance alongside standard deviations (Table 4).

As anticipated, all models score significantly lower in the zero-shot LSC setting than in-domain, with both metrics exhibiting similar trends. Training on Sem-Lex transfers better to unseen LSC handshapes than training on PopSign, particularly for the MLP (11.8% vs. 27.3% Hs.) and SHuBERT (13.9% vs. 18.3% Hs.) models. For SL-GCN, the two training corpora yield indistinguishable zero-shot performance (8.3% vs. 8.6% Hs.). Notably, while the MLP is the weakest model in-domain, it emerges as the *strongest* zero-shot model on LSC, leading both transfer metrics (62.6% Ph., 27.3% Hs.). We hypothesize that high-capacity spatio-temporal models overfit to source-dataset biases and pose extraction artifacts.

Mitigating the recording-format gap As outlined in Section 4.3, we re-evaluated each model under recording conditions closer to its training distribution to isolate cross-lingual transfer from recording-format disparities. To maintain a strict zero-shot source-only setup, harmonization is determined by the training corpus alone: each model is evaluated with spatial cropping matching its capture geometry and the frame rate closest to its training median—webcam at 25 fps for Sem-Lex-

Train	Model	ASL test		LSC zero-shot	
		Ph. (%) $\uparrow$	Hs. (%) $\uparrow$	Ph. (%) $\uparrow$	Hs. (%) $\uparrow$
PopSign	MLP	76.5 $\pm$ 0.1	55.4 $\pm$ 0.1	53.6 $\pm$ 1.6	11.8 $\pm$ 1.7
	SL-GCN	98.2$\pm$0.1	96.5$\pm$0.1	52.3 $\pm$ 0.7	8.3 $\pm$ 2.4
	SHuBERT	94.3 $\pm$ 0.1	89.7 $\pm$ 0.3	56.7$\pm$1.5	13.9$\pm$0.7
Sem-Lex	MLP	75.2 $\pm$ 0.2	48.8 $\pm$ 0.2	62.6$\pm$1.7	27.3$\pm$1.2
	SL-GCN	90.0$\pm$0.1	78.4$\pm$0.3	47.9 $\pm$ 2.6	8.6 $\pm$ 3.7
	SHuBERT	81.7 $\pm$ 0.2	64.9 $\pm$ 0.2	59.9 $\pm$ 1.3	18.3 $\pm$ 4.2

Table 4: In-domain ASL vs. zero-shot LSC transfer performance. Three model families are trained on PopSign and Sem-Lex, then evaluated on their respective ASL test splits and zero-shot on the LSC benchmark. Ph.: mean accuracy across handshape phonological features; Hs.: expected handshape accuracy. Cells report the mean $\pm$ standard deviation across random seeds. **Bold** indicates the best-performing model within each training corpus block.

trained models and phone at 25 fps for PopSign-trained ones (shaded cells in Table 5). Under this corpus-matched setting, the Sem-Lex MLP reaches 80.0% Ph. / 54.5% Hs., the Sem-Lex SHuBERT 65.6% / 22.1%, and the PopSign SL-GCN 70.4% / 29.5%—gains of +17.4 / +27.2, +5.7 / +3.8, and +18.1 / +21.2 points (Ph. / Hs.), respectively, over their uncropped 50 fps baselines (Table 4).

For completeness and as an ablation of both factors, Table 5 also reports performance across the full grid of crop and frame-rate combinations. Spatial cropping drives the vast majority of the gains: compared to uncropped baselines at matched frame rates, webcam and phone cropping yield average improvements of +11.1 / +11.3 and +10.5 / +10.6 (Ph. / Hs.), respectively, whereas frame-rate reduction offers marginal benefits (≤ 0.7 points averaged across models; see Table 13 in Appendix D).

Crucially, tuning the configuration on the target benchmark—rather than adhering to a strict zero-shot protocol—yields minimal gain. The grid optimum coincides with the corpus-matched setting for the strongest model (Sem-Lex MLP). For the remaining models, the grid optima exceed the corpus-matched configuration by at most 2.4 Ph. / 2.7 Hs. points (e.g., Sem-Lex SHuBERT at webcam, 50 fps: 66.1% / 23.5%; PopSign SL-GCN at phone, 16.7 fps: 72.5% / 31.5%). Thus, the observed improvements hold under a strictly zero-shot, source-only setup.

Appendix E decomposes these results per phonological feature: the corpus-matched harmonization gains concentrate on selected fingers and spread, while flexion remains the weakest feature.

	Model	Crop	Ph. (%) $\uparrow$			Hs. (%) $\uparrow$		
			50	25	16.7	50	25	16.7
PopSign	MLP	no crop	53.6	50.4	51.9	11.8	10.4	12.0
		webcam	71.9	72.6	71.9	25.3	26.9	27.8
		phone	70.7	70.2	69.5	26.0	26.4	25.9
	SL-GCN	no crop	52.3	55.0	54.7	8.3	11.8	9.9
		webcam	67.0	68.2	69.3	19.8	22.4	20.7
		phone	70.4	70.4	72.5	26.0	29.5	31.5
	SHuBERT	no crop	56.7	55.6	54.8	13.9	13.4	14.5
		webcam	55.6	56.2	56.8	14.6	15.8	15.1
		phone	55.1	56.9	56.4	14.9	15.8	13.5
Sem-Lex	MLP	no crop	62.6	60.5	62.9	27.3	22.4	25.0
		webcam	79.7	80.0	79.8	53.8	54.5	52.9
		phone	76.5	76.9	76.3	42.7	43.7	42.8
	SL-GCN	no crop	47.9	50.2	51.6	8.6	9.6	10.3
		webcam	56.3	57.2	57.8	13.1	14.5	15.9
		phone	54.3	58.2	59.1	13.1	17.2	15.2
	SHuBERT	no crop	59.9	58.5	58.2	18.3	15.7	16.3
		webcam	66.1	65.6	65.0	23.5	22.1	23.5
		phone	64.6	63.9	63.6	22.7	20.2	22.5

Table 5: Zero-shot evaluation on the LSC benchmark of video-crop (none, 4:3 webcam, 3:4 phone) and frame-rate (50, 25, 16.7) ablations for models trained on PopSign and Sem-Lex. Ph.: mean accuracy of five handshape phonological features; Hs.: expected handshape accuracy. **Bold** indicates the best Ph. and Hs. result per model configuration. Shaded cells mark the corpus-matched harmonization configuration—crop matching the training corpus, 25 fps (nearest to the training median)—fixed without consulting LSC results.

Target phonological heads To ablate the effect of auxiliary supervision, we retrained each model family on both ASL corpora to predict only the five handshape phonological features alongside the handshape class, rather than all 16 phonological heads. Holding model architectures and random seeds fixed, we present these comparative results in Table 15 (Appendix F). Removing auxiliary supervision yields no systematic benefits. In-domain, the shallow MLP is largely unaffected (≤ 0.3 percentage point shift), while deeper architectures suffer minor-to-moderate declines in handshape accuracy, particularly on Sem-Lex (SHuBERT -4.8 , SL-GCN -3.1). On zero-shot LSC transfer, performance fluctuates directionlessly (e.g., SHuBERT Ph. +2.9 vs. SL-GCN Ph. -4.7 on Sem-Lex) with overlapping seed standard deviations in nearly all cases. We conclude that while auxiliary heads offer mild regularization for deep models in-domain, they neither systematically aid nor degrade cross-lingual transfer.

	Ablation	ASL test		LSC zero-shot	
		Ph. (%) $\uparrow$	Hs. (%) $\uparrow$	Ph. (%) $\uparrow$	Hs. (%) $\uparrow$
PopSign	all channels (reference)	94.3$\pm$0.1	89.7$\pm$0.3	56.7$\pm$1.5	13.9$\pm$0.7
	hands-only	94.3$\pm$0.1	89.4 $\pm$ 0.3	56.2 $\pm$ 2.0	13.9$\pm$3.8
	body-only	55.4 $\pm$ 0.0	12.3 $\pm$ 0.0	50.3 $\pm$ 0.0 ††	2.7 $\pm$ 0.0 ††
	no pre-training	90.6 $\pm$ 0.5	83.1 $\pm$ 0.9	50.6 $\pm$ 4.6	8.6 $\pm$ 3.5
Sem-Lex	all channels (reference)	81.7 $\pm$ 0.2	64.9$\pm$0.2	59.9 $\pm$ 1.3	18.3 $\pm$ 4.2
	hands-only	81.8$\pm$0.1	64.2 $\pm$ 0.6	64.4$\pm$3.2	23.5$\pm$5.8
	body-only	55.0 $\pm$ 0.0	11.0 $\pm$ 1.1	50.3 $\pm$ 0.0 ††	2.7 $\pm$ 0.0 ††
	no pre-training	74.3 $\pm$ 0.3	50.7 $\pm$ 1.0	58.4 $\pm$ 1.4	14.9 $\pm$ 1.5

Table 6: Input channel and pre-training ablation for SHuBERT fine-tuned on PopSign and Sem-Lex, evaluated in-domain (ASL) and zero-shot (LSC). Ph.: mean phonological feature accuracy; Hs.: expected handshape accuracy. Cells report mean $\pm$ std over seeds; **bold** denotes best result per training corpus. †† Degenerate runs emitting constant majority-class predictions.

SHuBERT ablations We ablate SHuBERT along two axes (input channels and pretraining), varying one factor at a time against the all-channel pretrained reference (Table 6). **Input channels:** SHuBERT fuses four positional streams: face, left hand, right hand, and body posture. We evaluate configurations retaining all four channels, the two hand streams only, and the body stream only. Dropped channels are masked using SHuBERT’s pretrained mask embedding. Restricting the input to hand channels matches the all-channel reference with performance gaps in-domain falling within the standard deviation, demonstrating that face and body pose contribute minimally to handshape recognition. Conversely, the body-only model collapses: without hand signals, it predicts the training majority class, yielding a constant-predictor floor (marked †† in Table 6). **Pretraining:** training SHuBERT from scratch without self-supervised pretraining confirms that SSL pretraining provides substantial performance gains across all setups. On Sem-Lex, pretraining boosts in-domain phonological feature macro accuracy by +7.4 points and handshape accuracy by +14.2 points, whereas on PopSign the corresponding gains are +3.7 and +6.6 points. Consequently, zero-shot transfer to LSC exhibits a corresponding degradation when removing pre-training.

6 Discussion

Cross-Lingual Transfer and Phonological Universality Our findings show that zero-shot cross-lingual transfer from ASL to LSC is viable once domain gaps in recording format such as spatial cropping are explicitly harmonized. This supports

the hypothesis that phonological features capture foundational articulatory properties that transcend individual sign languages. However, important scope limitations remain. Our evaluation is restricted to a single historically related language pair (ASL $\rightarrow$ LSC) and a benchmark containing two videos per handshape from a single signer across 37 handshape classes (out of more than 43 described in LSC, excluding fingerspelling, numerals, and classifiers). In particular, signer-independent generalization within LSC cannot be established from this benchmark, and robustness to uncontrolled target recording conditions remains untested. Although our decoding framework remains mathematically agnostic to the target set $\mathcal{P}_{\text{LSC}}$, future work will evaluate this approach across larger multi-signer datasets and typologically unrelated sign language families to test cross-lingual universality.

Model Capacity, Dataset Biases, and Practical Efficiency A key insight from our experiments is the counterintuitive strength of the shallow MLP, which outperforms deeper architectures in zero-shot transfer. We hypothesize that high-capacity spatio-temporal models (such as SHuBERT and SL-GCN) overfit to source-dataset biases and pose-extraction artifacts. Because the MLP lacks temporal mechanics and receives only dominant-hand features, its capacity bottleneck prevents it from learning non-transferable contextual cues, yielding superior zero-shot generalization. Crucially, our ablations show that SHuBERT can be stripped to hand-only input streams and supervised on only six handshape-relevant heads without sacrificing accuracy. This stream pruning significantly reduces the annotation burden and computational overhead required to deploy phonological recognizers for handshape prediction in new target languages.

Static Phonological Representation Although the 5-dimensional phonological feature vector successfully grounds categorical decoding, its formulation reveals structural limits as observed by [Sehyr et al. \(2021\)](#). First, the feature mapping is non-injective: the five phonological categories used in this work fail to capture unselected finger flexion, causing distinct handshapes to map to identical phonological feature vectors. Second, reducing signs to a single static handshape simplifies dynamic signing reality, overlooking handshape transitions over time as well as non-dominant hand configurations. Extending the shared feature space to parameterize unselected fingers, dynamic tempo-

ral trajectories, and two-handed signs to the feature space would expand cross-lingual transfer across the complete manual phonological spectrum.

Expected Accuracy and Decoding Ties These representational constraints are directly manifested during evaluation, defining the expected handshape accuracy as a diagnostic metric rather than a standard classification performance score. Because candidate ties are resolved uniformly at random, even an ideal feature predictor achieves a theoretical ceiling of 94.6% (reflecting 35 unique canonical feature configurations across 37 target classes). Such ties stem from two distinct mechanisms: *structural ties* between handshape class pairs sharing identical phonological feature vectors, and *distance ties*, where a prediction lies equidistant from multiple valid LSC handshape classes. Future work can resolve distance ties by leveraging the model’s per-feature confidences—ensuring a single, deterministic prediction—and disambiguate structural ties only through supplementary information, such as unselected finger flexion features.

7 Conclusions

In this work, we introduced the first zero-shot cross-lingual framework for sign language handshape recognition, bridging American Sign Language (ASL) and Catalan Sign Language (LSC) through a shared, linguistically grounded phonological feature space. By decomposing handshapes into five phonological features and mapping predictions via a composite distance metric, our method enables direct target-language classification without requiring target-language phonological training labels.

Evaluated across three distinct model architectures (MLP, SL-GCN, and SHuBERT) and two source corpora (PopSign and Sem-Lex), our experiments on a single-signer LSC benchmark show that zero-shot transfer is viable—achieving up to 80.0% phonological feature accuracy and 54.5% expected handshape accuracy. Crucially, we show that isolating true cross-lingual transfer requires first harmonizing recording-format disparities such as spatial cropping. Our experiments reveal that low-capacity models (MLP) offer superior zero-shot generalization, providing evidence that model complexity does not necessarily translate into better cross-lingual transfer; we hypothesize that high-capacity models overfit source-dataset spatio-temporal biases. Complex multimodal transformers (SHuBERT), meanwhile, can be pruned to hand-only

streams without loss of handshape performance. These results support the hypothesis that phonological decomposition can provide a language-agnostic representation bridge: for the language pair studied, this work substantially reduces target-language annotation requirements and opens new avenues for scaling sign language technologies to underserved sign communities and low-resource sign languages worldwide.

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References

- Robbin Battison. 1978. *Lexical Borrowing in American Sign Language*. Linstok Press, Silver Spring, MD.
- Yunus Can Bilge, Nazli Ikizler-Cinbis, and Ramazan Gokberk Cinbis. 2024. [Cross-lingual few-shot sign language recognition](#). *Pattern Recognition*, 151:110374.
- Danielle Bragg, Oscar Koller, Mary Bellard, Larwan Berke, Patrick Boudreault, Annelies Braffort, Naomi Caselli, Matt Huenerfauth, Hernisa Kacorri, Tessa Verhoef, et al. 2019. Sign language recognition, generation, and translation: An interdisciplinary perspective. In *Proceedings of the 21st international ACM*

- SIGACCESS conference on computers and accessibility*, pages 16–31.
- Diane Brentari. 1998. *A Prosodic Model of Sign Language Phonology*. MIT Press, Cambridge, MA.
- Necati Cihan Camgoz, Simon Hadfield, Oscar Koller, Hermann Ney, and Richard Bowden. 2018. Neural sign language translation. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7784–7793. Ieee.
- Necati Cihan Camgoz, Oscar Koller, Simon Hadfield, and Richard Bowden. 2020. Sign language transformers: Joint end-to-end sign language recognition and translation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10023–10033.
- Alessa Carbo and Eric Nalisnick. 2025. [Improving handshape representations for sign language processing: A graph neural network approach](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 29122–29135, Suzhou, China. Association for Computational Linguistics.
- Naomi K Caselli, Zed Sevcikova Sehyr, Ariel M Cohen-Goldberg, and Karen Emmorey. 2017. ASL-LEX: A lexical database of american sign language. *Behavior research methods*, 49(2):784–801.
- Ashley Chow, Glenn Cameron, Mark Sherwood, Phil Culliton, Sam Sepah, Sohler Dane, and Thad Starner. 2023. [Google - Isolated Sign Language Recognition](#).
- Jens Forster, Christoph Schmidt, Thomas Hoyoux, Oscar Koller, Uwe Zelle, Justus H Piater, and Hermann Ney. 2012. Rwth-phoenix-weather: A large vocabulary sign language recognition and translation corpus. In *LREC*, volume 9, pages 3785–3789.
- Jens Forster, Christoph Schmidt, Oscar Koller, Martin Bellgardt, and Hermann Ney. 2014. Extensions of the sign language recognition and translation corpus rwth-phoenix-weather. In *LREC*, pages 1911–1916.
- Shester Gueuwou, Xiaodan Du, Greg Shakhnarovich, Karen Livescu, and Alexander H. Liu. 2025. [SHuBERT: Self-supervised sign language representation learning via multi-stream cluster prediction](#). *Preprint*, arXiv:2411.16765.
- Julie A. Hochgesang, Onno Crasborn, and Diane Lillo-Martin. 2019. [ASL Signbank](#). New Haven, CT: Haskins Lab, Yale University.
- Jundai Inoue, Daisuke Hara, and Makoto Miwa. 2026. [A resource and evaluation method for phonological continuity in Japanese Sign Language](#). In *Proceedings of the Fifteenth Language Resources and Evaluation Conference*, pages 9514–9524, Palma de Mallorca, Spain. ELRA Language Resource Association.
- Institut d’Estudis Catalans. 2025. [Corpus de referència de la llengua de signes catalana \(LSC\) \(CORPUS LSC\)](#). <https://corpuslsc.iec.cat/>.
- Songyao Jiang, Bin Sun, Lichen Wang, Yue Bai, Kunpeng Li, and Yun Fu. 2021. Skeleton aware multi-modal sign language recognition. In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 3408–3418. IEEE.
- Lee Kezar, Jesse Thomason, Naomi Caselli, Zed Sehyr, and Elana Pontecorvo. 2023. The sem-lex benchmark: Modeling asl signs and their phonemes. In *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility*, pages 1–10.
- Oscar Koller, Hermann Ney, and Richard Bowden. 2016. Deep hand: How to train a cnn on 1 million hand images when your data is continuous and weakly labelled. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3793–3802.
- Kezhou Lin, Xiaohan Wang, Linchao Zhu, Ke Sun, Bang Zhang, and Yi Yang. 2023. Gloss-free end-to-end sign language translation. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12904–12916.
- Camillo Lugaresi, Jiuqiang Tang, Hadon Nash, Chris McClanahan, Esha Uboweja, Michael Hays, Fan Zhang, Chuo-Ling Chang, Ming Guang Yong, Juhyun Lee, et al. 2019. Mediapipe: A framework for building perception pipelines. *arXiv preprint arXiv:1906.08172*.
- Mathias Müller, Zifan Jiang, Amit Moryossef, Annette Rios Gonzales, and Sarah Ebling. 2023. Considerations for meaningful sign language machine translation based on glosses. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 682–693.
- Alexandra Navarrete-González. 2020. [Phonology: 1. sublexical structure](#). In Josep Quer and Gemma Barberà, editors, *A Grammar of Catalan Sign Language (LSC)*, 1st edition, SIGN-HUB Sign Language Grammar Series.
- Josep Quer and Gemma Barberà, editors. 2020. *A Grammar of Catalan Sign Language (LSC)*, 1st edition. SIGN-HUB Sign Language Grammar Series.
- Zed Sevcikova Sehyr, Naomi Caselli, Ariel M Cohen-Goldberg, and Karen Emmorey. 2021. The ASL-LEX 2.0 project: A database of lexical and phonological properties for 2,723 signs in american sign language. *The Journal of Deaf Studies and Deaf Education*, 26(2):263–277.
- Thad Starner, Sean Forbes, Matthew So, David Martin, Rohit Sridhar, Gururaj Deshpande, Sam Sepah, Sahir Shahryar, Khushi Bhardwaj, Tyler Kwok, et al. 2023. Popsign asl v1. 0: An isolated american sign language dataset collected via smartphones. *Advances in Neural Information Processing Systems*, 36:184–196.

- William C. Stokoe. 1960. Sign language structure: An outline of the visual communication systems of the American deaf. *Studies in Linguistics, Occasional Papers*, 8.
- Garrett Tanzer and Biao Zhang. 2025. Youtube-sl-25: A large-scale, open-domain multilingual sign language parallel corpus. In *International Conference on Learning Representations*, volume 2025, pages 81921–81934.
- Sandrine Tornay, Marzieh Razavi, and Mathew Magimai Doss. 2020. Towards multilingual sign language recognition. In *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 6309–6313. IEEE.
- Dave Uthus, Garrett Tanzer, and Manfred Georg. 2023. Youtube-asl: A large-scale, open-domain american sign language-english parallel corpus. *Advances in Neural Information Processing Systems*, 36:29029–29047.
- Henri Wittmann. 1991. Classification linguistique des langues signées non vocalement. *Revue québécoise de linguistique théorique et appliquée*, 10(1):215–288.
- World Federation of the Deaf. 2026. [Faq: Deaf communities and sign languages](#). Accessed: 2026-08-21.
- Benjia Zhou, Zhigang Chen, Albert Clapés, Jun Wan, Yanyan Liang, Sergio Escalera, Zhen Lei, and Du Zhang. 2023. Gloss-free sign language translation: Improving from visual-language pretraining. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 20814–20824. IEEE.

A LSC Handshape Inventory and Feature Mapping

LSC handshapes were derived from the [SignHub LSC Grammar: Finger Configuration](#). We inferred selected fingers from the finger configuration table, relying either on explicit mentions in the handshape names or on standard number and letter conventions. Configuration values correspond to the original header row, while spread was manually annotated following ASL-LEX rules.

Table 7 presents the complete inventory of 43 LSC handshapes alongside their original SignHub annotations and mapped ASL-LEX features. The deterministic alignment procedure used to link these two feature spaces is detailed in Section 4.2.1. Note that spread distinguishes class 5 (spread) from *B* (unspread) exclusively for *extended* and *curved open* configurations. The remaining configurations collapse into a single *5B* class. We encode this merged class as unspread based on SignHub’s LSC finger configuration illustrations and the benchmark clips.

As noted by [Sehyr et al. \(2021\)](#) and discussed in Section 6, the five ASL-LEX handshape features—*selected fingers*, *flexion*, *spread*, *thumb position*, and *thumb contact*—cannot distinguish all handshapes, as certain classes differ exclusively in the flexion of unselected fingers.

LSC Handshape	SignHub-inferred LSC annotations			ASL-LEX 2.0 phonological features				
	Sel. fingers	Configuration	Spread	Sel. fingers	Flexion	Spread	Thumb pos.	Thumb cont.
6-extended	t	extended	NA	t	FullyOpen	NA	Open	NA
6-curved_open	t	curved open	NA	t	Curved/Bent	NA	Open	NA
6-closed	t	closed	NA	t	FullyClosed	NA	Open	NA
1-extended	i	extended	NA	i	FullyOpen	NA	Closed	0
1-curved_open	i	curved open	NA	i	Curved/Bent	NA	Closed	0
1-closed	i	closed	NA	i	FullyClosed	NA	Closed	1
middle-extended	m	extended	NA	m	FullyOpen	NA	Closed	0
middle-flat_open	m	flat open	NA	m	Flat	NA	Closed	0
i-extended	p	extended	NA	p	FullyOpen	NA	Closed	0
i-curved_open*	p	curved open	NA	p	Curved/Bent	NA	Closed	0
7-extended	ti	extended	NA	i	FullyOpen	NA	Open	0
7-flat_open [†]	ti	flat open	NA	i	Flat	NA	Open	0
7-flat_closed [‡]	ti	flat closed	NA	i	Flat	NA	Open	1
7-curved_open	ti	curved open	NA	i	Curved/Bent	NA	Open	0
index+thumb-flat_open [†]	ti	flat open	NA	i	Flat	NA	Open	0
index+thumb-flat_closed [‡]	ti	flat closed	NA	i	Flat	NA	Open	1
index+thumb-curved_closed	ti	curved closed	NA	i	Curved/Bent	NA	Open	1
middle+thumb-flat_open*	tm	flat open	NA	m	Flat	NA	Open	0
middle+thumb-flat_closed*	tm	flat closed	NA	m	Flat	NA	Open	1
middle+thumb-curved_closed*	tm	curved closed	NA	m	Curved/Bent	NA	Open	1
2-extended	im	extended	spread	im	FullyOpen	1	Closed	0
2-curved_open	im	curved open	spread	im	Curved/Bent	1	Closed	0
2-curved_closed	im	curved closed	spread	im	Curved/Bent	1	Closed	1
n-extended	im	extended	unspread	im	FullyOpen	0	Closed	0
n-curved_open*	im	curved open	unspread	im	Curved/Bent	0	Closed	0
Y-extended	tp	extended	NA	p	FullyOpen	NA	Open	0
u-extended	ip	extended	NA	ip	FullyOpen	NA	Closed	0
8-extended	tim	extended	spread	im	FullyOpen	1	Open	0
8-flat_closed	tim	flat closed	spread	im	Flat	0	Open	1
8-curved_open	tim	curved open	spread	im	Curved/Bent	1	Open	0
4-extended	imrp	extended	spread	imrp	FullyOpen	1	Closed	0
4-flat_open	imrp	flat open	spread	imrp	Flat	0	Closed	0
5-extended	timrp	extended	spread	imrp	FullyOpen	1	Open	0
5-curved_open	timrp	curved open	spread	imrp	Curved/Bent	1	Open	0
b-extended	timrp	extended	unspread	imrp	FullyOpen	0	Open	0
b-curved_open	timrp	curved open	unspread	imrp	Curved/Bent	0	Open	0
5B-flat_open	timrp	flat open	unspread**	imrp	Flat	0	Open	0
5B-flat_closed	timrp	flat closed	unspread**	imrp	Flat	0	Open	1
5B-curved_closed	timrp	curved closed	unspread**	imrp	Curved/Bent	0	Open	1
5B-closed	timrp	closed	unspread**	imrp	FullyClosed	NA	Open	1
pinkie+middle+thumb-extended	tmp	extended	NA	mp	FullyOpen	NA	Open	0
3-extended	imr	extended	spread	imr	FullyOpen	1	Closed	0
3-curved_open*	imr	curved open	spread	imr	Curved/Bent	1	Closed	0

Table 7: Mapping of the 43 LSC handshape classes ([Navarrete-González, 2020](#)) to ASL-LEX 2.0 phonological features, alongside their SignHub-inferred annotations. Asterisks (*) mark the six handshape classes lacking visual clips in the benchmark. The double asterisk (**) denotes collapsed *5B* classes encoded as unspread (see text). Classes sharing a dagger ([†], [‡]) possess identical ASL-LEX feature configurations, differing only in the flexion of unselected fingers.

B Phonological Feature Inventories Across Datasets

To analyze the structural granularity of the feature spaces across datasets, Table 8 contrasts the class counts across each phonological feature for Sem-Lex, PopSign, and the LSC Benchmark. Table 9 provides an itemized comparison of the value inventories for the five primary handshape features, explicitly denoting merged classes (e.g., *Curved/Bent*) and structural gaps between ASL-LEX and LSC.

Comparing these label spaces highlights a key structural divergence in *Selected Fingers*: the combination *mp* (middle and pinkie) is attested in LSC (e.g., in the *pinkie+middle+thumb-extended* handshape) but absent from the ASL-LEX training set. Consequently, our model cannot directly predict the exact *mp* value during zero-shot evaluation. Nevertheless, because target handshape retrieval relies on a distance metric over the combined feature space, *pinkie+middle+thumb-extended* can still be successfully recovered within the candidate set. To systematically handle unobserved finger combinations in future work, finger selection could be decomposed into independent per-digit binary features rather than treating joint configurations as discrete atomic classes.

Phonological Feature	Sem-Lex	PopSign	LSC Benchmark
<i>Other phonological features</i>			
Major Location	5	5	–
Minor Location	37	25	–
Second Minor Location	37	16	–
Contact	2	2	–
Sign Type	6	5	–
Repeated Movement	2	2	–
Path Movement	8	7	–
Wrist Twist	2	2	–
Spread Change	3	3	–
Nondominant Handshape	56	25	–
<i>Handshape phonological features</i>			
Selected Fingers	12	9	9
Flexion	8	7	4
Spread	3	3	3
Thumb Position	2	2	2
Thumb Contact	3	3	3
<i>Handshape class</i>			
Handshape	58	37	37

Table 8: Number of classes per phonological feature in each dataset’s label space. Sem-Lex uses the full ASL-LEX 2.0 coding; PopSign’s space is reduced to the values attested among its 250 glosses; the LSC Benchmark column is the canonical target space of the five handshape features in ASL-LEX format (Curved and Bent flexion merged) plus the 37 handshape classes – the remaining features have no LSC annotation (–). Counts include the NA class where the annotation has structurally-absent values (Table 9 lists the value inventories of the five handshape features).

Feature	ASL-LEX 2.0 values	LSC values (ASL-LEX format)
Selected Fingers	t, i, m, r , p, im, ip, mr , imr, imp , mrp , imrp	t, i, m, p, im, ip, mp , imr, imrp
Flexion	FullyOpen, Flat, Curved , Bent , FullyClosed, Stacked , Crossed , NA	FullyOpen, Flat, Curved/Bent , Fully-Closed
Spread	0, 1, NA	0, 1, NA
Thumb Position	Open, Closed	Open, Closed
Thumb Contact	0, 1, NA	0, 1, NA

Table 9: Value inventories of the five phonological features strictly related to handshape: the ASL-LEX 2.0 space the models are trained on vs the reduced LSC target space in ASL-LEX format. **Bold** marks the changes: ASL-LEX values with no identical LSC counterpart (Crossed, Stacked, and flexion NA have no LSC realisation; Curved and Bent collapse into the merged Curved/Bent class; four selected-finger combinations are unattested in LSC) and LSC values absent from the ASL-LEX space (the merged Curved/Bent; mp, attested in LSC but not in ASL-LEX). NA is a real class, marking structurally-absent values, never missing supervision.

Feature	Test Top-1 Accuracy (%) $\uparrow$				
	Ref.	Released Poses		Re-posed	
		Ours	Δ	Ours	Δ
<i>Gloss ISR (Tab. 4/6): the feature heads' encoder</i>					
Gloss top-1	66.6	57.1	-9.5	72.2	+5.6
Gloss top-3	81.5	71.1	-10.4	85.6	+4.1
<i>Phonological features (Tab. 5, multitask)</i>					
Major Location	87.5	79.4	-8.1	88.4	+0.9
Minor Location	78.1	67.3	-10.8	80.0	+1.9
2nd Minor Location	77.2	69.2	-8.0	80.4	+3.2
Contact	88.6	85.1	-3.5	90.7	+2.1
Sign Type	87.9	80.8	-7.1	88.8	+0.9
Repeated Movement	85.4	81.5	-3.9	88.6	+3.2
Path Movement	75.4	69.7	-5.7	80.4	+5.0
Wrist Twist	92.6	89.5	-3.1	93.5	+0.9
Spread Change	89.5	84.3	-5.2	91.2	+1.7
Nondominant Handshape	81.7	72.4	-9.3	82.9	+1.2
Selected Fingers	90.2	84.6	-5.6	92.3	+2.1
Flexion	81.0	73.8	-7.2	83.3	+2.3
Spread	88.0	81.7	-6.3	89.6	+1.6
Thumb Position	91.5	87.7	-3.8	92.6	+1.1
Thumb Contact	91.1	86.4	-4.7	92.2	+1.1
Handshape	74.7	65.6	-9.1	78.4	+3.7
Average	85.0	78.7	-6.3	87.1	+2.1

Table 10: SL-GCN reproduction fidelity on the test split. We compare the results reported by Sem-Lex (Kezar et al., 2023) against our reproduction runs. Our models are evaluated on two pose sets: the original *Released Poses* provided by the benchmark, and *Re-extracted*, which denotes the poses we independently extracted from the same videos. The Δ columns indicate the absolute percentage point difference between our reproduction and the baseline.

C Reproduction Results

To ensure the reliability of our baseline implementations, we reproduce experiments on both SL-GCN (Kezar et al., 2023) and SHuBERT (Gueuwou et al., 2025). Table 10 reports the reproduction fidelity for SL-GCN across two distinct pose inputs: the official pose files shipped with the Sem-Lex release versus pose sequences extracted directly from the raw dataset videos using our pipeline. Evaluating on the official released poses reveals a systematic performance drop (averaging -6.3 percentage points), whereas our re-extracted pose sequences reliably match and slightly exceed original paper metrics ($+2.1$ percentage points on average across phonological tasks).

Table 11 evaluates our fine-tuned SHuBERT baseline against the published Sem-Lex test results. Our reproduced SHuBERT model consistently matches or marginally outperforms published per-feature Recall@1 scores in all 16 phonological dimensions, achieving an average performance gain of $+0.86$ percentage points.

Phonological Feature	Test Recall@1 (%) $\uparrow$		
	Ref.	Ours	Δ
Major Location	84.77	85.03	+0.26
Minor Location	71.30	71.73	+0.43
2nd Minor Location	73.28	73.64	+0.36
Contact	86.84	87.35	+0.51
Sign Type	84.64	84.66	+0.02
Repeated Movement	82.65	84.01	+1.36
Path Movement	72.75	72.92	+0.17
Wrist Twist	90.58	91.48	+0.90
Spread Change	81.60	82.85	+1.25
Nondominant Handshape	76.32	76.72	+0.40
Selected Fingers	79.53	81.04	+1.51
Flexion	72.64	74.38	+1.74
Spread	79.42	80.38	+0.96
Thumb Position	86.04	87.03	+0.99
Thumb Contact	84.74	85.76	+1.02
Handshape	62.93	64.89	+1.96
Average	79.38	80.24	+0.86

Table 11: SHuBERT reproduction fidelity evaluated on the Sem-Lex test set. We compare the per-feature Recall@1 results reported in the original SHuBERT baseline (Gueuwou et al., 2025) against our fine-tuned reproduction. The Δ column indicates the absolute percentage point difference between our reproduction and the baseline.

D Further results for mitigation of recording-format gap

To provide a full accounting of stability and individual factor contributions in our domain-adaptation experiments, this section presents detailed breakdowns of the cropping format and frame-rate ablations. Table 12 extends Table 5 from the main text by reporting sample standard deviations across three random seeds for every cell in the 3×3 grid of crop configurations (uncropped, 4:3 webcam, and 3:4 phone portrait) and frame rates (50 fps, 25 fps, and 16.7 fps). Table 13 isolates the marginal treatment effects (Δ) of spatial cropping relative to uncropped inputs, as well as temporal frame-rate reductions relative to native 50 fps video.

Spatial alignment drives gains. Spatial cropping yields substantial performance improvements for all models ($+10.5\%$ – $+11.3\%$ average gain), peaking at $+28.9\%$ for MLP handshape accuracy on Sem-Lex. This shows pose-based models are sensitive to video framing, though SHuBERT is less affected by uncropped inputs. Conversely, frame-rate reductions (from 50 fps to 25 or 16.7 fps) have a negligible impact ($\leq +0.7\%$ average change). This confirms that spatial domain shift—rather than temporal resolution—is the primary bottleneck in zero-shot cross-dataset transfer.

		Ph. (%) $\uparrow$			Hs. (%) $\uparrow$			
Model	Crop	50 fps	25 fps	16.7 fps	50 fps	25 fps	16.7 fps	
PopSign	MLP	no crop	53.6 $\pm$ 1.6	50.4 $\pm$ 2.6	51.9 $\pm$ 2.1	11.8 $\pm$ 1.7	10.4 $\pm$ 2.8	12.0 $\pm$ 2.9
		webcam crop	71.9 $\pm$ 2.1	72.6$\pm$2.0	71.9 $\pm$ 2.1	25.3 $\pm$ 2.0	26.9 $\pm$ 2.8	27.8$\pm$3.7
		phone crop	70.7 $\pm$ 2.3	70.2 $\pm$ 1.1	69.5 $\pm$ 1.3	26.0 $\pm$ 1.9	26.4 $\pm$ 1.8	25.9 $\pm$ 1.4
	SL-GCN	no crop	52.3 $\pm$ 0.7	55.0 $\pm$ 1.0	54.7 $\pm$ 2.3	8.3 $\pm$ 2.4	11.8 $\pm$ 0.5	9.9 $\pm$ 1.7
		webcam crop	67.0 $\pm$ 1.4	68.2 $\pm$ 1.8	69.3 $\pm$ 0.7	19.8 $\pm$ 2.2	22.4 $\pm$ 3.3	20.7 $\pm$ 2.2
		phone crop	70.4 $\pm$ 1.1	70.4 $\pm$ 0.9	72.5$\pm$0.4	26.0 $\pm$ 1.6	29.5 $\pm$ 2.7	31.5$\pm$1.0
	SHuBERT	no crop	56.7 $\pm$ 1.5	55.6 $\pm$ 1.0	54.8 $\pm$ 2.3	13.9 $\pm$ 0.7	13.4 $\pm$ 1.1	14.5 $\pm$ 1.6
		webcam crop	55.6 $\pm$ 1.5	56.2 $\pm$ 1.5	56.8 $\pm$ 1.1	14.6 $\pm$ 1.4	15.8 $\pm$ 1.0	15.1 $\pm$ 0.7
		phone crop	55.1 $\pm$ 1.4	56.9$\pm$1.3	56.4 $\pm$ 1.2	14.9 $\pm$ 0.2	15.8$\pm$1.6	13.5 $\pm$ 1.8
Sem-Lex	MLP	no crop	62.6 $\pm$ 1.7	60.5 $\pm$ 1.9	62.9 $\pm$ 1.6	27.3 $\pm$ 1.2	22.4 $\pm$ 0.8	25.0 $\pm$ 1.2
		webcam crop	79.7 $\pm$ 1.2	80.0$\pm$0.9	79.8 $\pm$ 0.7	53.8 $\pm$ 2.6	54.5$\pm$3.6	52.9 $\pm$ 3.3
		phone crop	76.5 $\pm$ 1.2	76.9 $\pm$ 1.1	76.3 $\pm$ 0.4	42.7 $\pm$ 1.9	43.7 $\pm$ 2.2	42.8 $\pm$ 2.3
	SL-GCN	no crop	47.9 $\pm$ 2.6	50.2 $\pm$ 3.1	51.6 $\pm$ 4.4	8.6 $\pm$ 3.7	9.6 $\pm$ 2.1	10.3 $\pm$ 4.0
		webcam crop	56.3 $\pm$ 4.3	57.2 $\pm$ 3.3	57.8 $\pm$ 0.9	13.1 $\pm$ 4.8	14.5 $\pm$ 1.2	15.9 $\pm$ 0.5
		phone crop	54.3 $\pm$ 5.3	58.2 $\pm$ 5.3	59.1$\pm$1.0	13.1 $\pm$ 3.6	17.2$\pm$4.4	15.2 $\pm$ 1.7
	SHuBERT	no crop	59.9 $\pm$ 1.3	58.5 $\pm$ 0.7	58.2 $\pm$ 1.8	18.3 $\pm$ 4.2	15.7 $\pm$ 3.2	16.3 $\pm$ 4.6
		webcam crop	66.1$\pm$2.3	65.6 $\pm$ 2.0	65.0 $\pm$ 1.7	23.5$\pm$5.3	22.1 $\pm$ 1.8	23.5 $\pm$ 5.1
		phone crop	64.6 $\pm$ 0.7	63.9 $\pm$ 1.3	63.6 $\pm$ 1.9	22.7 $\pm$ 2.5	20.2 $\pm$ 1.1	22.5 $\pm$ 3.6

Table 12: Video-crop $\times$ frame-rate ablation on the LSC benchmark with seed spreads: the appendix sibling of Table 5. Models are trained on PopSign and Sem-Lex and evaluated zero-shot on the LSC Benchmark. Ph. is the mean accuracy over the five handshape phonological features; Hs. is the expected handshape accuracy; decoding is constrained to the LSC inventory. The *Crop* column gives the type of image cropping applied to the LSC clips (none, webcam-style 4:3, or phone-style 3:4 portrait), with each model’s input re-extracted from the cropped video; the numeric sub-columns give the video frame rate in fps (native 50, subsampled 25 and 16.7). **Bold** marks, for each trained model, its best Ph. and best Hs. cell over the whole grid; shaded cells mark the corpus-matched harmonization configuration (crop matching the training corpus, 25 fps), fixed without consulting LSC results. Cells are mean $\pm$ sample sd over three seeds.

		Phonological feature macro Δ (%) $\uparrow$				Expected handshape Δ (%) $\uparrow$			
Model		webcam	phone	25	16.7	webcam	phone	25	16.7
PopSign	MLP	+20.2$\pm$2.3	+18.2 $\pm$ 2.0	-1.0 $\pm$ 0.4	-1.0 $\pm$ 0.3	+15.3$\pm$3.6	+14.7 $\pm$ 3.1	+0.2 $\pm$ 0.3	+0.9 $\pm$ 0.9
	SL-GCN	+14.2 $\pm$ 1.9	+17.1$\pm$1.2	+1.3 $\pm$ 1.3	+2.3 $\pm$ 1.0	+11.0 $\pm$ 2.8	+19.0$\pm$2.5	+3.2 $\pm$ 2.9	+2.7 $\pm$ 1.9
	SHuBERT	+0.5$\pm$0.8	+0.5$\pm$0.5	+0.5$\pm$0.3	+0.2 $\pm$ 0.4	+1.2$\pm$1.7	+0.8 $\pm$ 1.0	+0.5 $\pm$ 0.4	-0.1 $\pm$ 1.1
	mean	+11.6 $\pm$ 10.1	+11.9$\pm$9.9	+0.2 $\pm$ 1.2	+0.5 $\pm$ 1.7	+9.1 $\pm$ 7.2	+11.5$\pm$9.5	+1.3 $\pm$ 1.7	+1.2 $\pm$ 1.4
Sem-Lex	MLP	+17.8$\pm$1.2	+14.6 $\pm$ 1.1	-0.5 $\pm$ 0.4	+0.1 $\pm$ 0.4	+28.9$\pm$2.4	+18.2 $\pm$ 1.4	-1.0 $\pm$ 0.5	-1.0 $\pm$ 0.7
	SL-GCN	+7.2 $\pm$ 3.4	+7.3$\pm$4.7	+2.3 $\pm$ 1.9	+3.3 $\pm$ 2.1	+5.0 $\pm$ 3.1	+5.6$\pm$4.1	+2.2 $\pm$ 1.0	+2.2 $\pm$ 1.6
	SHuBERT	+6.7$\pm$2.2	+5.2 $\pm$ 2.3	-0.9 $\pm$ 1.1	-1.3 $\pm$ 1.0	+6.3$\pm$0.8	+5.1 $\pm$ 2.3	-2.2 $\pm$ 1.7	-0.7 $\pm$ 2.3
	mean	+10.6$\pm$6.3	+9.0 $\pm$ 4.9	+0.3 $\pm$ 1.8	+0.7 $\pm$ 2.4	+13.4$\pm$13.4	+9.6 $\pm$ 7.4	-0.3 $\pm$ 2.3	+0.1 $\pm$ 1.8
Mean	MLP	+19.0$\pm$1.7	+16.4 $\pm$ 2.5	-0.7 $\pm$ 0.4	-0.5 $\pm$ 0.7	+22.1$\pm$9.6	+16.4 $\pm$ 2.5	-0.4 $\pm$ 0.9	-0.1 $\pm$ 1.3
	SL-GCN	+10.7 $\pm$ 4.9	+12.2$\pm$6.9	+1.8 $\pm$ 0.7	+2.8 $\pm$ 0.7	+8.0 $\pm$ 4.2	+12.3$\pm$9.4	+2.7 $\pm$ 0.7	+2.4 $\pm$ 0.3
	SHuBERT	+3.6$\pm$4.4	+2.8 $\pm$ 3.3	-0.2 $\pm$ 1.0	-0.5 $\pm$ 1.0	+3.8$\pm$3.6	+2.9 $\pm$ 3.0	-0.8 $\pm$ 1.9	-0.4 $\pm$ 0.5
	mean	+11.1$\pm$7.5	+10.5 $\pm$ 7.2	+0.3 $\pm$ 1.3	+0.6 $\pm$ 1.8	+11.3$\pm$9.9	+10.6 $\pm$ 7.7	+0.5 $\pm$ 2.0	+0.7 $\pm$ 1.5

Table 13: Treatment effects of the crop $\times$ frame-rate grid, one row per trained model. *webcam* and *phone* are the crop’s effect against the uncropped clips, averaged over the three frame rates; 25 and 16.7 are the frame rate’s effect against native 50 fps, averaged over the three crops. *Mean* rows average the unrounded deltas of the rows in their scope (a corpus, a model family across both corpora, or all six models); their $\pm$ is the sd across the averaged rows, not a seed sd – every other $\pm$ is a sample sd over three seeds.

		Per-feature accuracy (%) $\uparrow$							
Model	Config	SF	Flex.	Sprd.	Th.Pos.	Th.Cnt.	Ph.		
Majority class		27.0	40.5	54.1	62.2	67.6	50.3		
PopSign	MLP	no crop, 50	47.7	44.6	64.9	64.4	46.4	53.6	
		phone, 25	76.6	60.8	87.8	60.8	64.9	70.2	
	SL-GCN	no crop, 50	49.5	40.1	53.6	64.9	53.2	52.3	
		phone, 25	74.3	53.6	77.5	66.7	79.7	70.4	
	SHuBERT	no crop, 50	39.6	44.1	64.9	67.6	67.1	56.7	
		phone, 25	44.1	44.1	63.5	67.6	65.3	56.9	
Sem-Lex	MLP	no crop, 50	64.0	48.6	63.5	74.8	62.2	62.6	
		webcam, 25	88.3	69.8	87.8	77.5	76.6	80.0	
	SL-GCN	no crop, 50	43.7	33.8	45.0	61.3	55.9	47.9	
		webcam, 25	57.7	41.9	58.6	63.1	64.9	57.2	
		SHuBERT	no crop, 50	49.1	47.7	64.9	69.4	68.5	59.9
			webcam, 25	57.2	49.5	68.0	75.7	77.5	65.6

Table 14: Per-feature error analysis on the LSC benchmark: accuracy of each of the five phonological features and their mean (Ph.), computed over the 74 benchmark clips and three seeds. Each model is evaluated zero-shot on the raw recordings (no crop, native 50 fps) and with corpus-matched harmonization (crop matching the training corpus, 25 fps; shaded in Table 5); the two settings differ only in label-free video pre-processing. Majority class: a constant predictor answering the benchmark’s most frequent value per feature. Bold: best config per model and feature. SF: selected fingers.

E Per-Feature Error Analysis

Table 14 decomposes the zero-shot LSC results into the five phonological features, reporting the accuracy of each feature together with their mean (Ph.). Each model is evaluated zero-shot on the raw recordings (no crop, 50 fps) and with its corpus-matched harmonization; the rows thus decompose exactly the aggregate values of Table 5. The *majority class* row reports the accuracy of a constant predictor answering the benchmark’s most frequent value per feature. Because our models are trained on ASL corpora whose class priors differ from the benchmark’s and never observe its distribution, this row is a reference line rather than an expected floor. Three patterns emerge. First, flexion—a four-way distinction of finger curvature—is the weakest or tied-weakest feature in every corpus-matched configuration. Second, harmonization gains concentrate on selected fingers and spread (e.g., +24.3 points each for the Sem-Lex MLP), while thumb position is nearly unaffected. Third, measured against the majority references, the learned signal is largest for selected fingers (88.3% vs. a 27.0% reference for the Sem-Lex MLP) and smallest for the two thumb features, whose accuracies stay within a few points of their references.

	Model	Setting	ASL Test		LSC Zero-Shot	
			Ph. (%) $\uparrow$	Hs. (%) $\uparrow$	Ph. (%) $\uparrow$	Hs. (%) $\uparrow$
PopSign	MLP	all16	76.5 $\pm$ 0.1	55.4 $\pm$ 0.1	53.6 $\pm$ 1.6	11.8 $\pm$ 1.7
		heads6	76.7$\pm$0.1	55.5$\pm$0.2	55.1$\pm$0.7	12.8$\pm$0.5
	SL-GCN	all16	98.2$\pm$0.1	96.5$\pm$0.1	52.3 $\pm$ 0.7	8.3$\pm$2.4
		heads6	98.0 $\pm$ 0.1	96.1 $\pm$ 0.1	53.9$\pm$1.1	5.4 $\pm$ 2.3
	SHuBERT	all16	94.3$\pm$0.1	89.7$\pm$0.3	56.7$\pm$1.5	13.9 $\pm$ 0.7
		heads6	93.7 $\pm$ 0.2	88.4 $\pm$ 0.5	55.5 $\pm$ 3.6	16.5$\pm$2.9
Sem-Lex	MLP	all16	75.2 $\pm$ 0.2	48.8$\pm$0.2	62.6$\pm$1.7	27.3$\pm$1.2
		heads6	75.5$\pm$0.1	48.8$\pm$0.2	59.1 $\pm$ 1.6	27.3$\pm$1.2
	SL-GCN	all16	90.0$\pm$0.1	78.4$\pm$0.3	47.9$\pm$2.6	8.6$\pm$3.7
		heads6	88.8 $\pm$ 0.3	75.3 $\pm$ 0.4	43.2 $\pm$ 4.9	5.0 $\pm$ 1.0
	SHuBERT	all16	81.7$\pm$0.2	64.9$\pm$0.2	59.9 $\pm$ 1.3	18.3 $\pm$ 4.2
		heads6	79.9 $\pm$ 0.2	60.1 $\pm$ 0.3	62.8$\pm$3.0	18.5$\pm$3.3

Table 15: Supervision head-set ablation across all three model families. Models are grouped by training set (PopSign and Sem-Lex blocks) and evaluated on two test sets: in-domain (ASL Test) and zero-shot (LSC Zero-Shot). Setting: full 16-head training (all16) vs. 6 handshape-relevant heads (heads6). Ph.: mean accuracy across the handshape phonological features; Hs.: expected handshape accuracy. Cells report mean $\pm$ standard deviation across seeds; **bold** denotes the higher value per paired ablation setting.

F Ablation on Target Phonological Heads

To evaluate the impact of auxiliary multi-task supervision, we compare models trained to predict all 16 phonological features (denoted as all16) against those trained exclusively on the 6 handshape-related heads, i.e. the five handshape phonological features alongside the handshape class, (denoted as heads6), as introduced in Section 5.

Table 15 details the empirical results, revealing a clear trade-off between in-domain performance and zero-shot transfer capability. For in-domain ASL evaluation, the full 16-head auxiliary supervision (all16) acts as a valuable multi-task regularizer for the more complex architectures (SL-GCN and SHuBERT), consistently yielding higher accuracy than heads6. However, in zero-shot LSC transfer, restricting supervision to heads6 frequently results in competitive or even superior performance. For example, Sem-Lex-trained SHuBERT improves from 59.9% to 62.8% (Ph.) when discarding the auxiliary non-handshape heads. Similarly, the simpler MLP baseline slightly benefits from the focused heads6 setting across both datasets. This suggests that while full multi-tasking helps robustly model in-domain ASL phonology, focusing solely on handshape variables can sometimes prevent over-specialization to ASL’s specific feature distributions, aiding cross-lingual generalization.